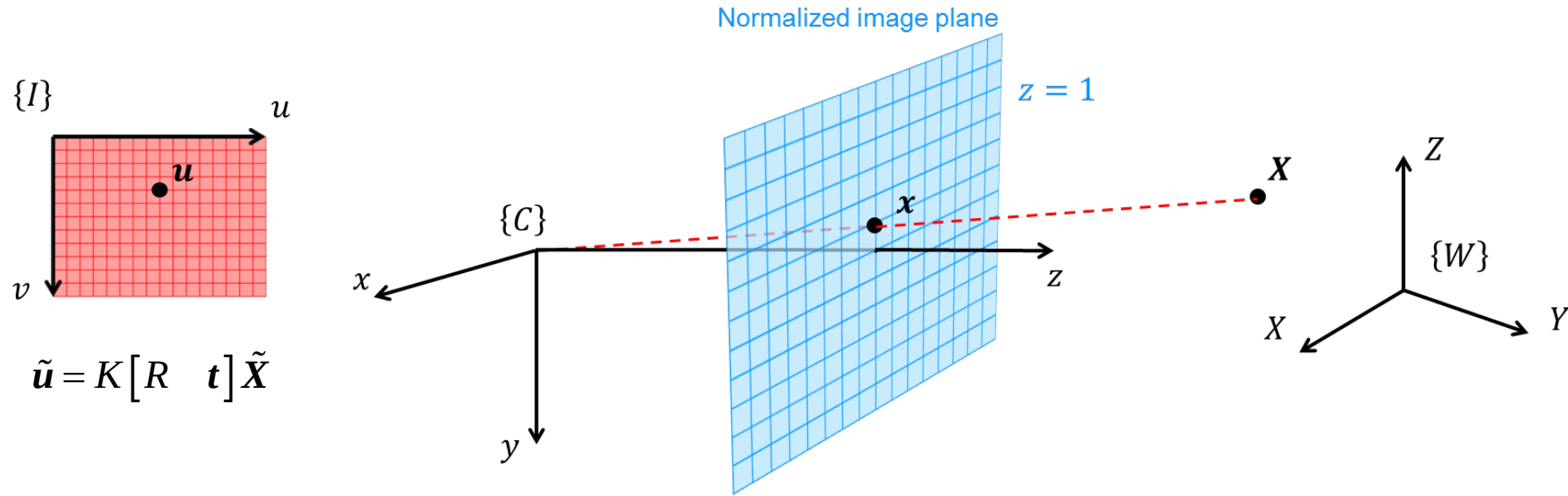


Lecture 5.1

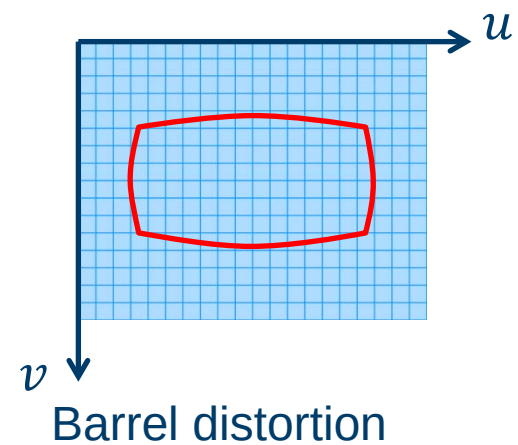
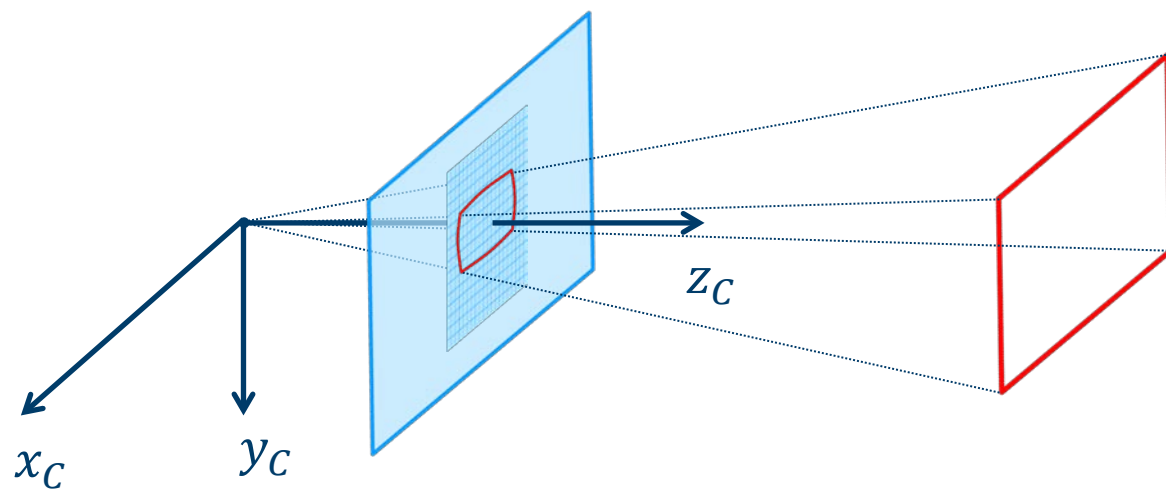
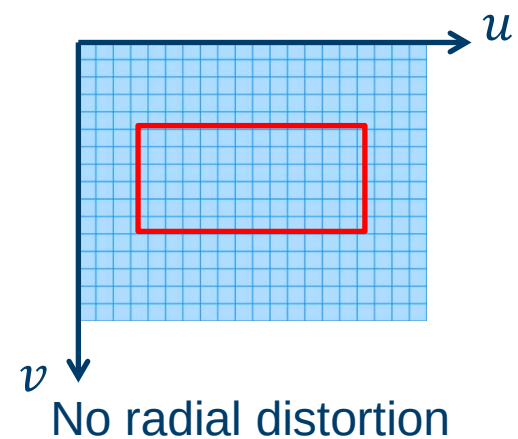
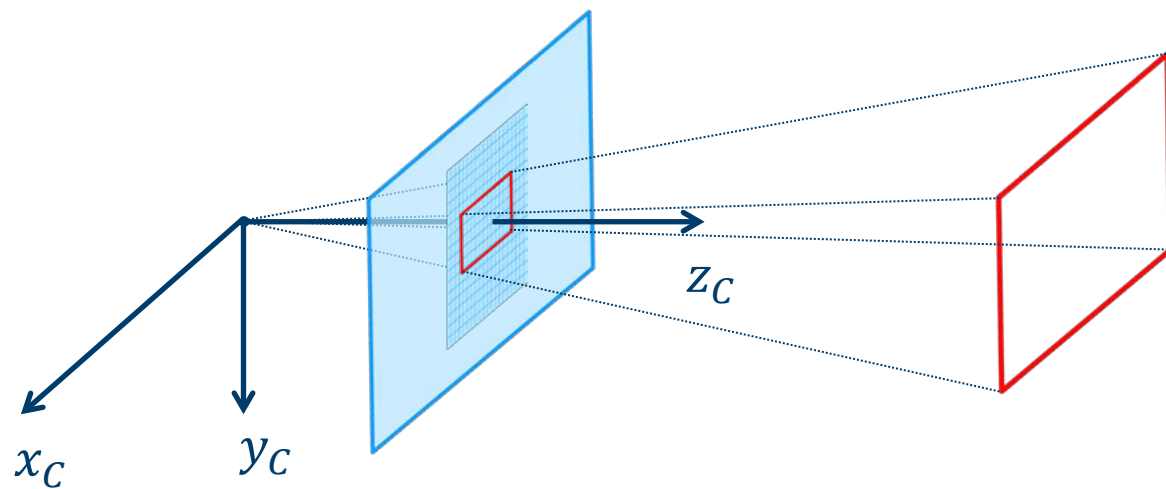
Camera calibration

Introduction

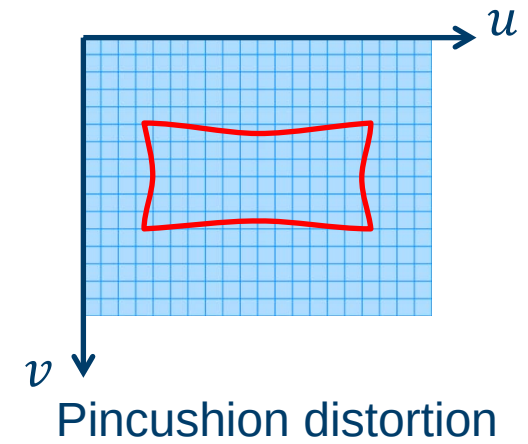
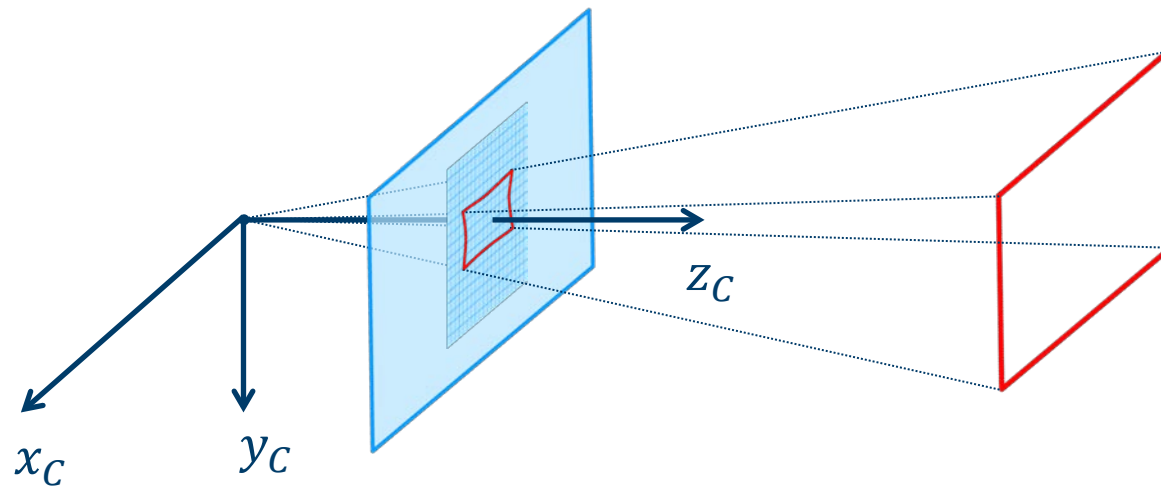
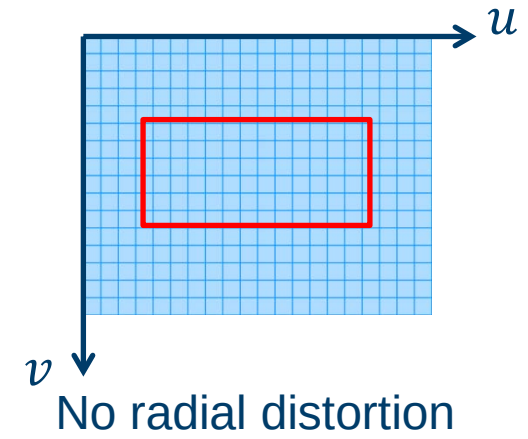
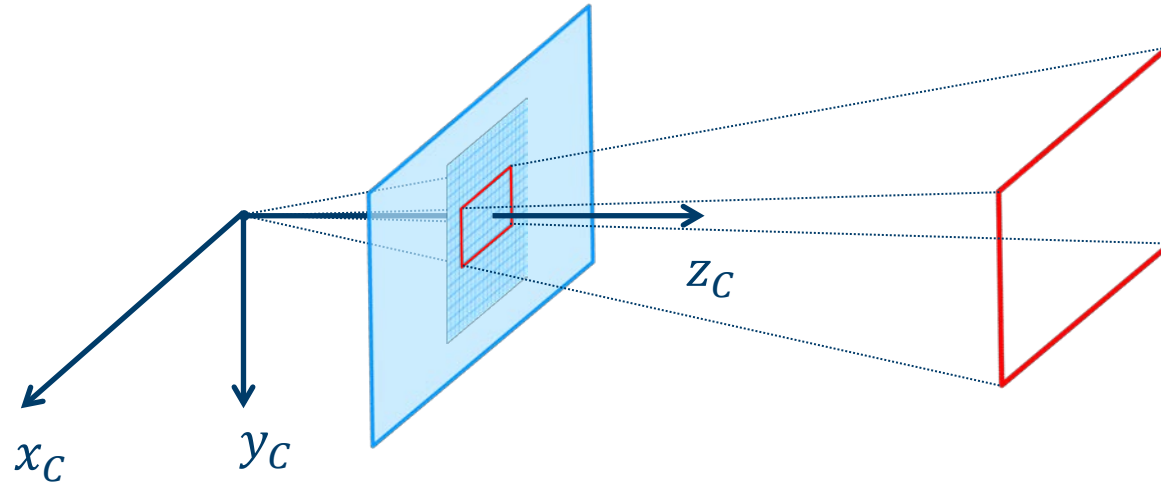


- The perspective camera model describes a 3D to 2D transformation that is consistent with the pinhole geometry
- No cameras fit this model perfectly – They all suffer from some kind of distortion
- If we want to use images for geometrical computations we need to take this distortion into account

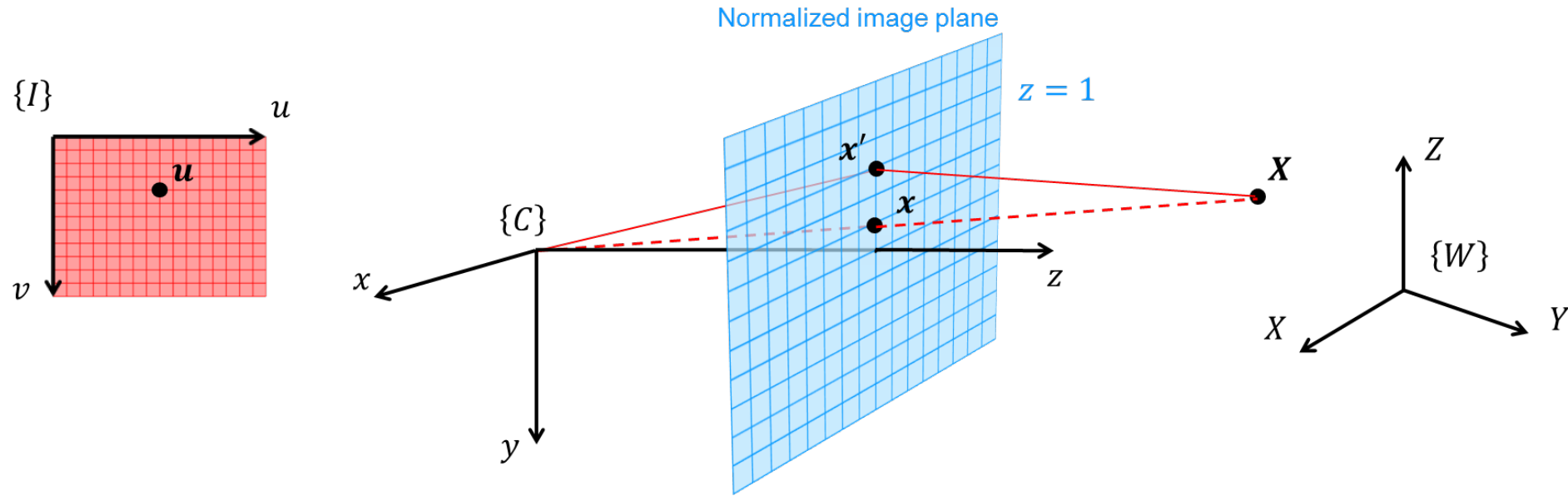
Introduction



Introduction



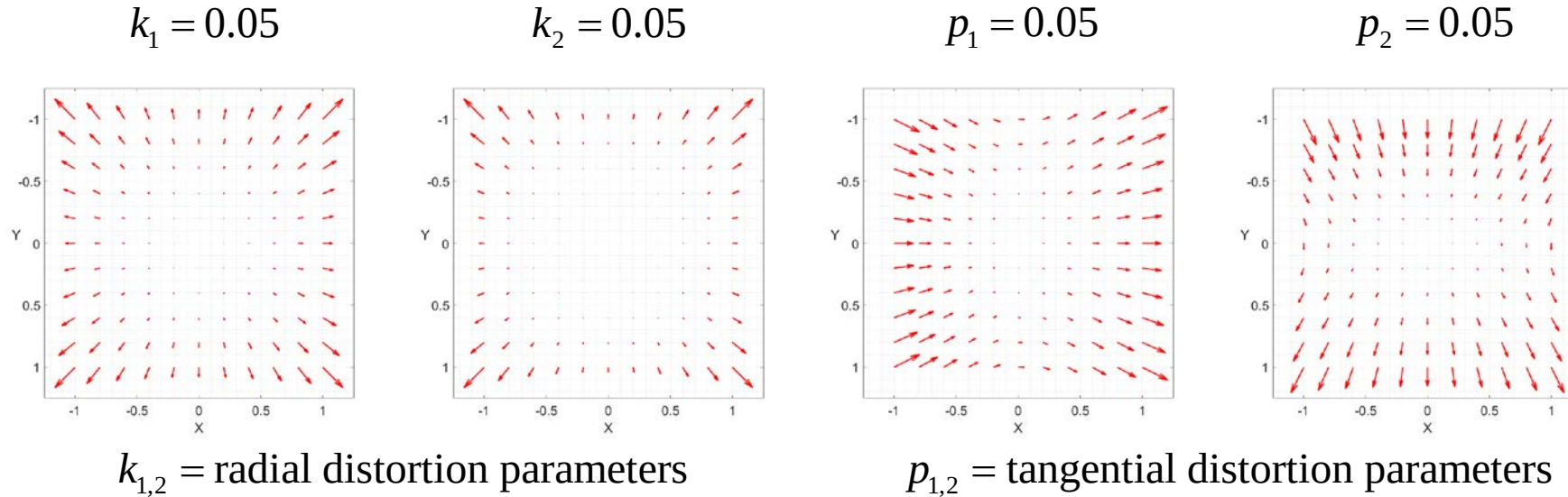
Undistortion



- A distortion model describes the relationship between undistorted coordinates x and distorted coordinates x' of the normalized image plane
- This example model describes both radial distortion and tangential distortion

$$\begin{aligned}
 x &= x' \left(1 + k_1 r^2 + k_2 r^4 \right) + 2p_1 x' y' + p_2 \left(r^2 + 2x'^2 \right) \\
 y &= y' \left(1 + k_1 r^2 + k_2 r^4 \right) + p_1 \left(r^2 + 2y'^2 \right) + 2p_2 x' y'
 \end{aligned}
 \quad \text{where } r^2 = x'^2 + y'^2$$

Undistortion

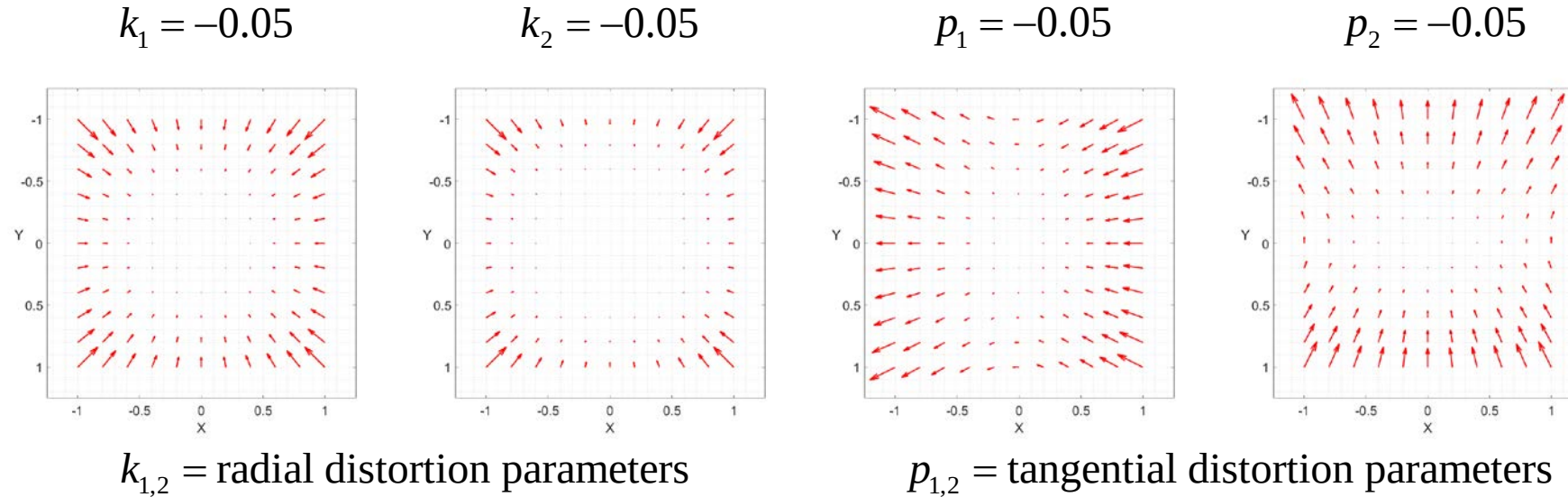


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where $r^2 = x'^2 + y'^2$

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where $r^2 = x'^2 + y'^2$

Undistortion

Original image



$$\tilde{\mathbf{u}} \neq K \begin{bmatrix} R & \mathbf{t} \end{bmatrix} \tilde{\mathbf{X}}$$

undistortion



Undistorted image

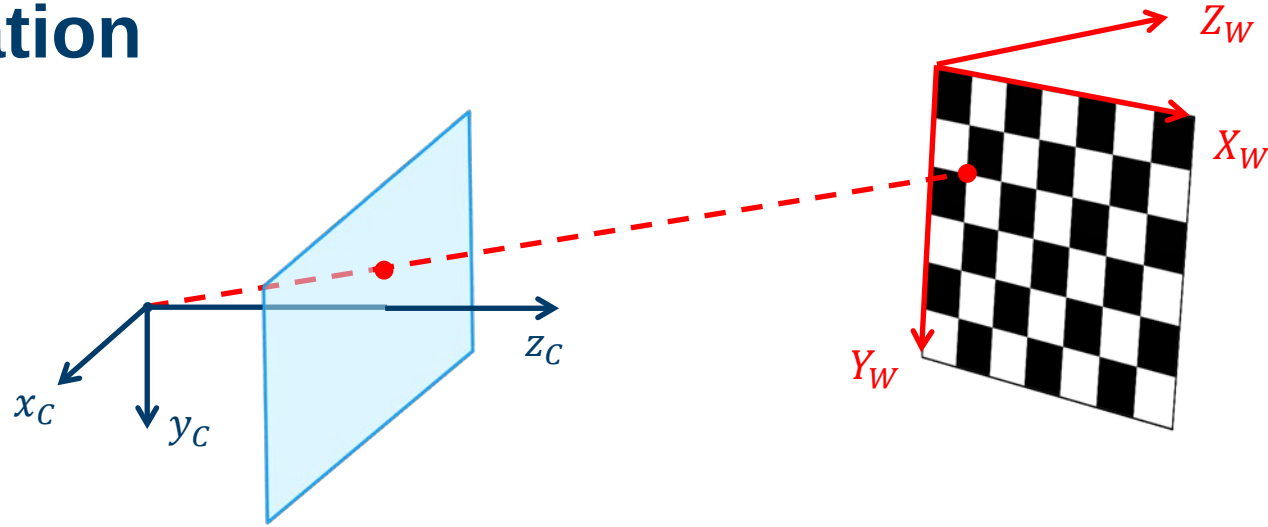


$$\tilde{\mathbf{u}} = K \begin{bmatrix} R & \mathbf{t} \end{bmatrix} \tilde{\mathbf{X}}$$

- We can use the distortion model to warp the original image into the so called undistorted image
- The undistorted image satisfy the perspective camera model and are thus well suited for geometrical computations
- Since the distortion model depends on K for the undistorted camera, it is custom to estimate both in a common calibration process

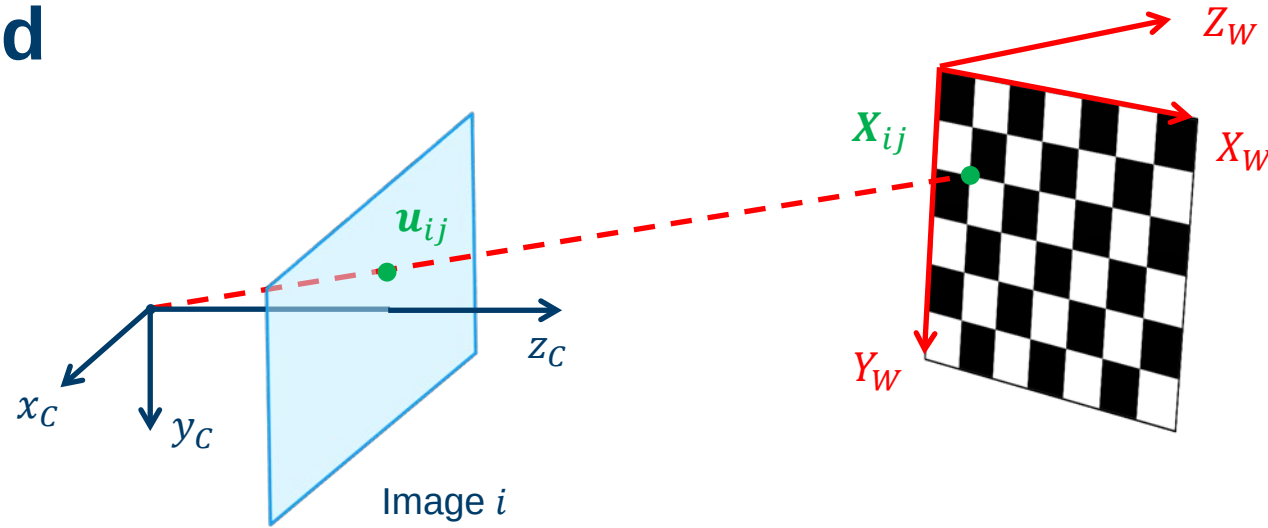
Images: <http://www.robots.ox.ac.uk/~vgg/hzbook/>

Camera calibration



- Camera calibration is a process where we estimate the intrinsic parameters f_u , f_v , s , c_u , c_v and distortion parameters for a camera
- One of the most commonly used calibration algorithms was described by Zhengyou Zhang in the paper "*A Flexible New Technique for Camera Calibration*" in 2000
- Zhang's method is based on using a planar calibration object, e.g. a chessboard where we know the size of the tiles

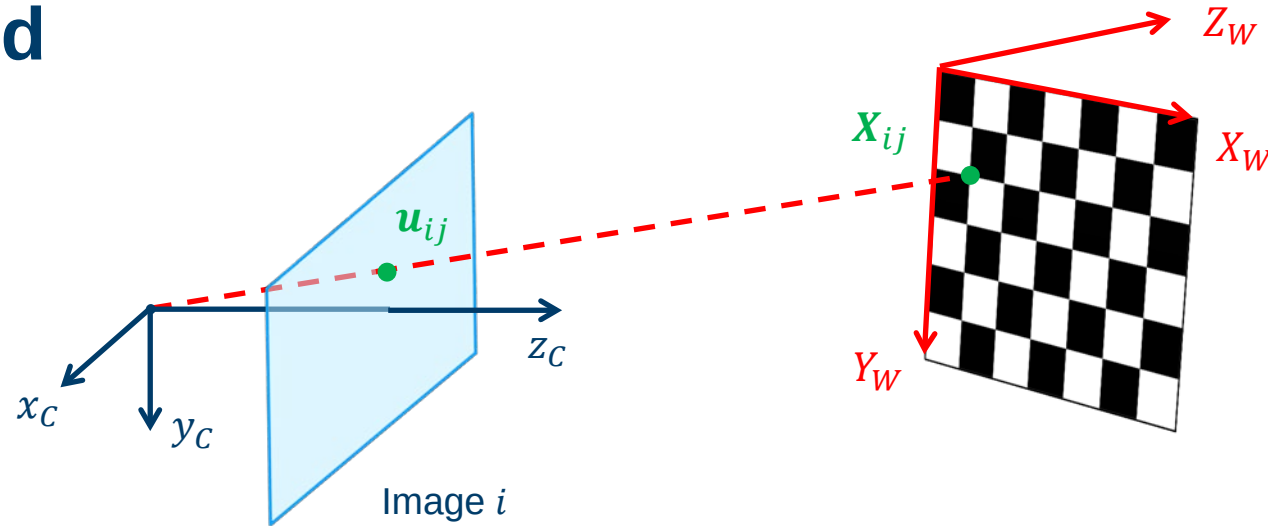
Zhang's method



1. Capture multiple images (at least 3) of the planar calibration object
2. For each image, estimate the homography H between the 2D surface of the calibration object and the image
3. Based on these homographies, estimate K (DLT) and from this all R_i 's and t_i 's
4. Use the estimated parameters as the starting point of an iterative non-linear optimization of the full set of intrinsic parameters (including distortion parameters κ)

$$\min_{K, \kappa, R_i, t_i} \sum_i \sum_j \left\| u_{ij} - \hat{u}(K, \kappa, R_i, t_i; X_{ij}) \right\|^2$$

Zhang's method

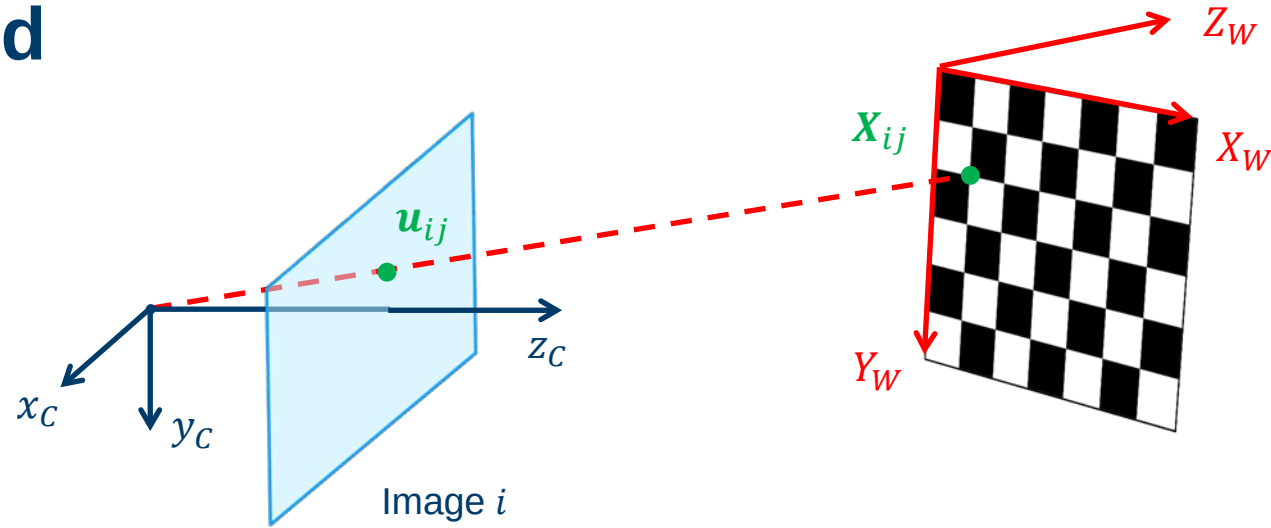


- Given that the calibration object is planar, the 3D-2D relationship between points on the calibration object and points in the image is described by a homography

$$\begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = K \underbrace{\begin{bmatrix} \mathbf{r}_1 & \mathbf{r}_2 & \mathbf{r}_3 & \mathbf{t} \end{bmatrix}}_{[R \quad t]} \begin{bmatrix} X \\ Y \\ 0 \\ 1 \end{bmatrix} \Rightarrow \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = K \begin{bmatrix} \mathbf{r}_1 & \mathbf{r}_2 & \mathbf{t} \end{bmatrix} \begin{bmatrix} X \\ Y \\ 1 \end{bmatrix} \Rightarrow \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = H \begin{bmatrix} X \\ Y \\ 1 \end{bmatrix}$$

- We can estimate the homography H from a minimum of 4 point correspondences between the image and the calibration object

Zhang's method



- Since $\mathbf{r}_1$ and $\mathbf{r}_2$ are columns of a rotation matrix, they impose 2 constraints on the elements of H

$$\left. \begin{array}{l} \mathbf{r}_1^T \mathbf{r}_2 = 0 \\ \mathbf{r}_1^T \mathbf{r}_1 = \mathbf{r}_2^T \mathbf{r}_2 \end{array} \right\} \Rightarrow \left\{ \begin{array}{l} \mathbf{h}_1^T K^{-T} K^{-1} \mathbf{h}_2 = 0 \\ \mathbf{h}_1^T K^{-T} K^{-1} \mathbf{h}_1 = \mathbf{h}_2^T K^{-T} K^{-1} \mathbf{h}_2 \end{array} \right.$$

- From N images we get N homographies, giving rise to $2N$ constraints allowing us to solve for the different elements of K using linear methods (DLT)

Summary



- Undistortion
 - For geometrical computations we work on undistorted images/feature points
- Calibration
 - Estimate K + distortion model parameters from images
 - Zhang's method
 - OpenCV, Matlab, Kalibr (<https://github.com/ethz-asl/kalibr>)

- Additional reading
 - Szeliski: 6.3
- Optional reading
 - *A flexible new technique for camera calibration*, by Z. Zhang